

Operation: addItem(itemTypeId: Integer, amount: Integer, expirationDate: LocalDate, damaged: Boolean, inWarehouse: Boolean)

Cross References: Use Cases: Update Inventory and Alert on Shortages

Preconditions: An ItemType instance with the matching itemTypeId exists in the system. The amount is greater than 0.

Postconditions:

- amount new Item instances were created (instance creation).
- The attributes (expirationDate, damaged, inWarehouse) of the new Item instances were initialized (attribute modification).
- An association was formed between each new Item instance and its corresponding ItemType (association formed).
- If the items were added to the warehouse (inWarehouse is true):

The warehouseQuantity attribute of the ItemType was increased by amount (attribute modification).

- If the items were added to the shelves (inWarehouse is false):

The shelfQuantity attribute of the ItemType was increased by amount (attribute modification).